

**DAI 1000**

# **Digital and AI Literacy**

**L3 & L4 - AI and Technology  
Convergence**

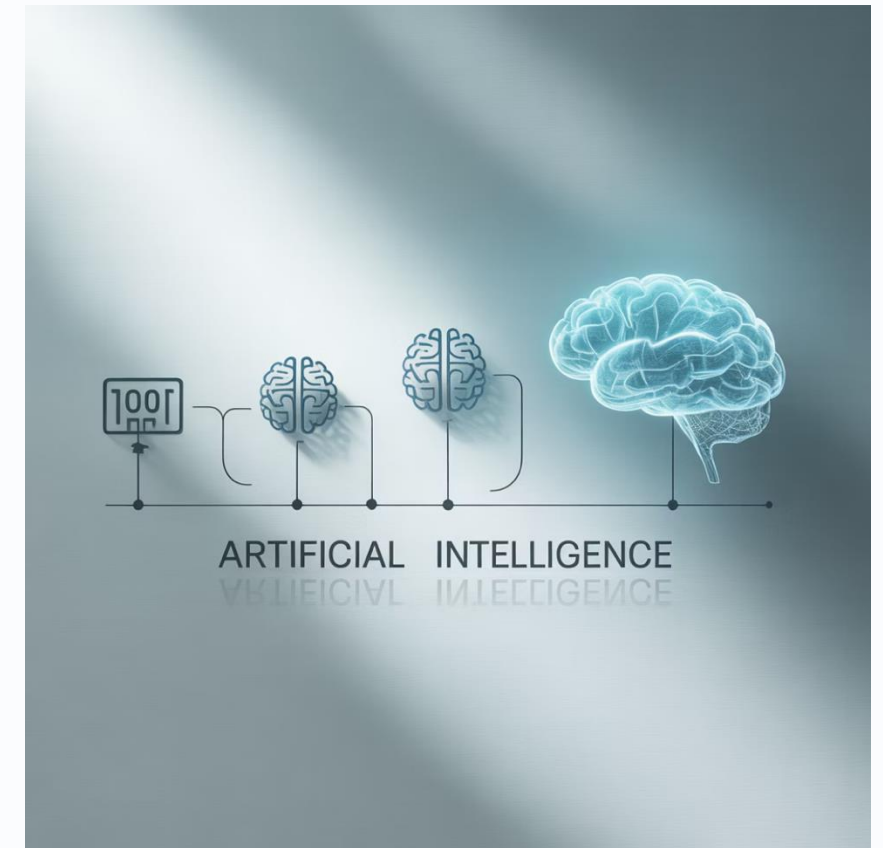
If you have the chance to define  
intelligence for a machine,  
what would it be?

[https://www.bilibili.com/video/BV1bc411g7P7/?spm\\_id\\_from=333.337.search-card.all.click&vd\\_source=b0e15bdbda6a17abcc8c653915e36a8f](https://www.bilibili.com/video/BV1bc411g7P7/?spm_id_from=333.337.search-card.all.click&vd_source=b0e15bdbda6a17abcc8c653915e36a8f)

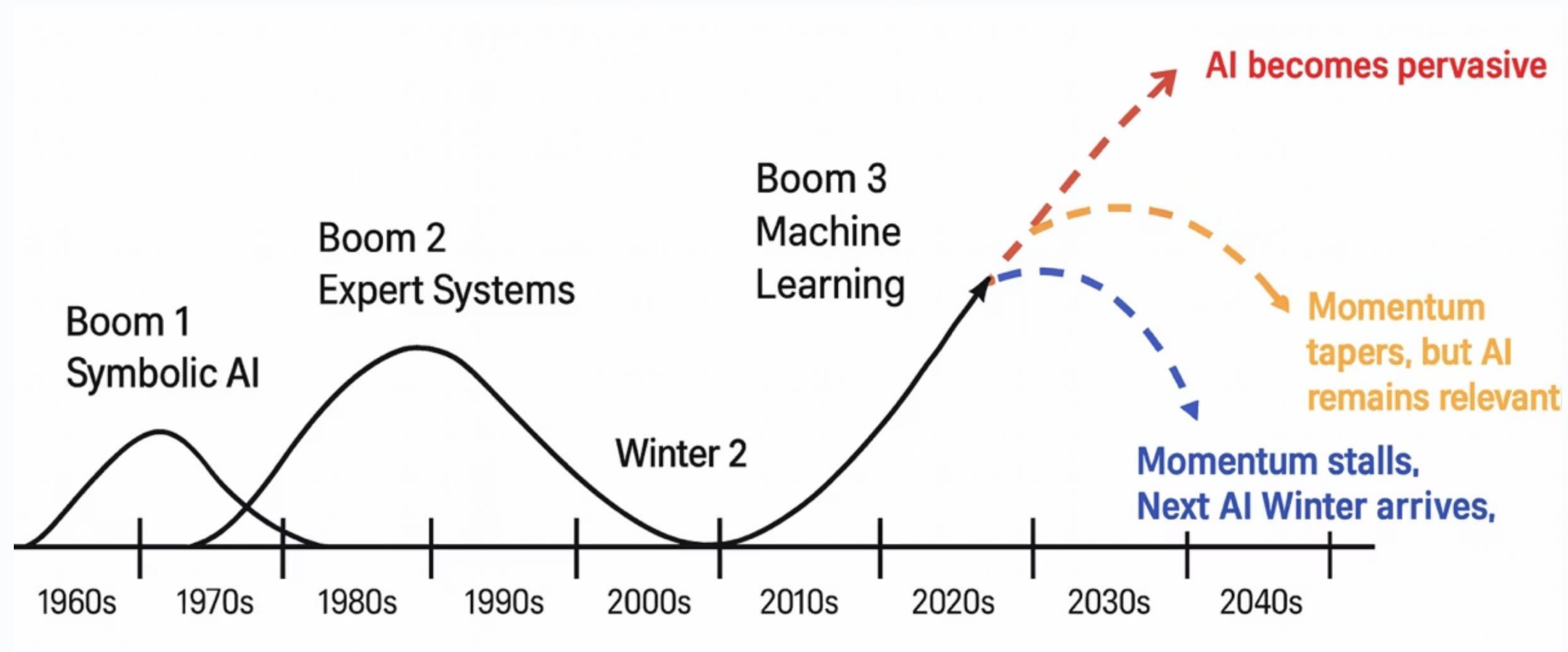
# Class Objectives

Let's start with where all this started

- AI history
  - Milestones
  - Two AI winters



# AI History



(may not be accurate in time, just to show the trend)

# AI Evolution Summary

| Period                                | Main Directions                                                                                    | Approaches                                                                                                | Main Issue                                                                                           |
|---------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Boom 1<br><br>(1950s– Late 60s)       | Can machines think like Human?                                                                     | Symbolic AI<br><br>Solve logic puzzles, play chess                                                        | Reality too complex for rules;<br>Not Achievable<br><br>(Resulting in Winter 1, Late 1960s–Late 70s) |
| Boom 2<br><br>(Late 1970s– Early 90s) | Can machines be experts?                                                                           | Use rules to diagnose or configure                                                                        | Fragile and hard to maintain<br><br>(Resulting in Winter 2, Early 1990s–Early 2000s)                 |
| Boom 3<br><br>( Mid 2000s–Now)        | Can machines learn from data?<br><br>Can machines perceive?<br><br>Can machines create and reason? | Recognize patterns in data<br><br>See, hear, understand language<br><br>Generate content, assist in tasks | Limited to narrow tasks                                                                              |

# Boom 1 (1950s– Late 60s): Can Machines Think Like Humans?

- Main Goal

Build a machine that can think and reason, like a human.
- What AI Could Do
  - Solve simple logic problems
  - Play games like checkers or chess using brute-force search
  - Parse very simple sentences in natural language
- Key Technologies

Symbolic AI (rules, logic), early neural nets (Perceptron)
- Major Issue

AI worked in toy problems, but failed in the real world. Everything had to be manually coded. It lacked common sense.
- What Failed
  - Overconfidence — people thought intelligence = logic
  - Early machine translation failed badly, leading to the First AI Winter

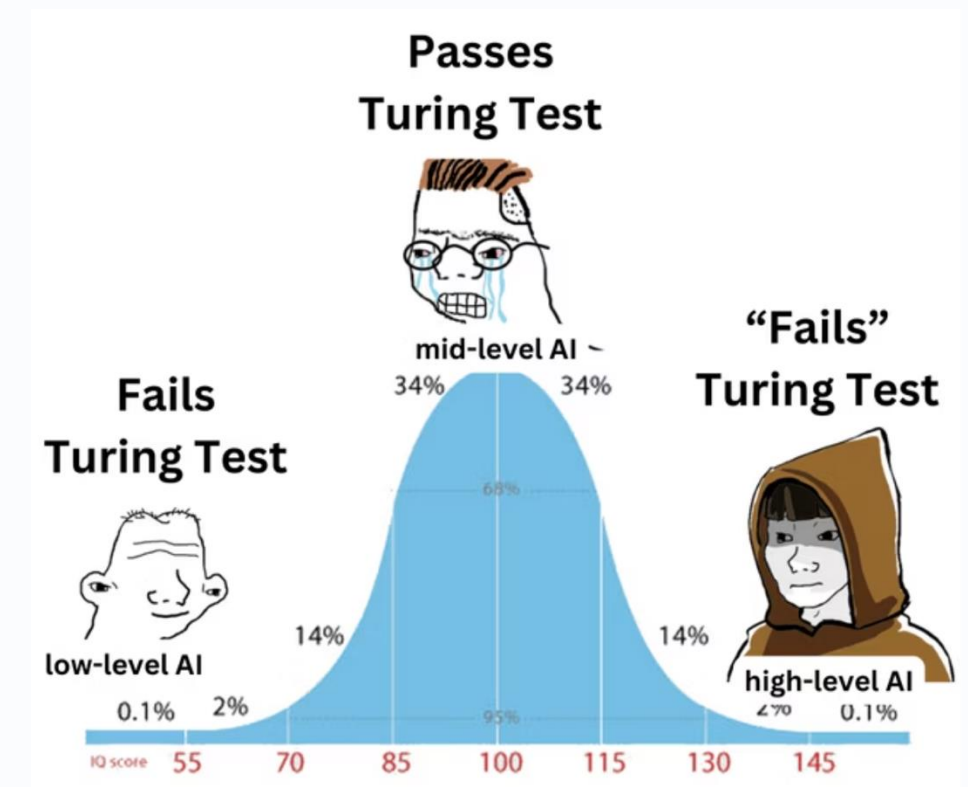
# A Review of AI History

- **1950:** Claude Shannon published "Programming a Computer for Playing Chess"
- The first published article on developing a chess-playing computer program

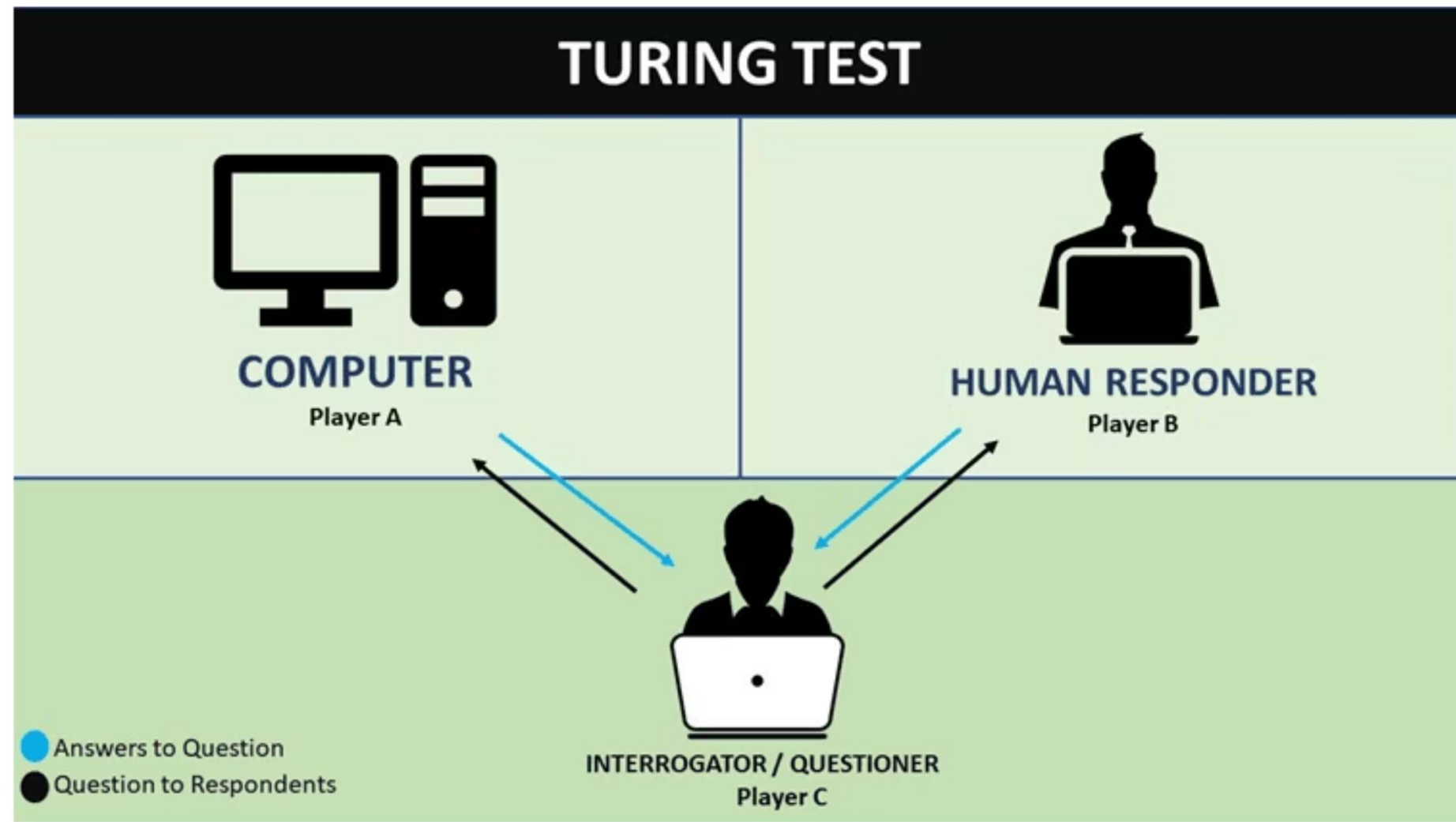


# A Review of AI History

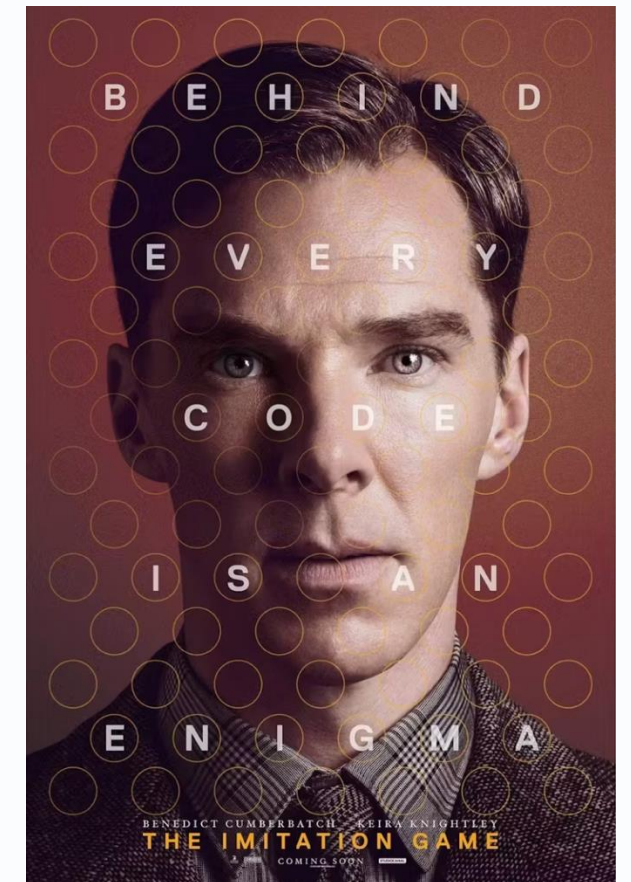
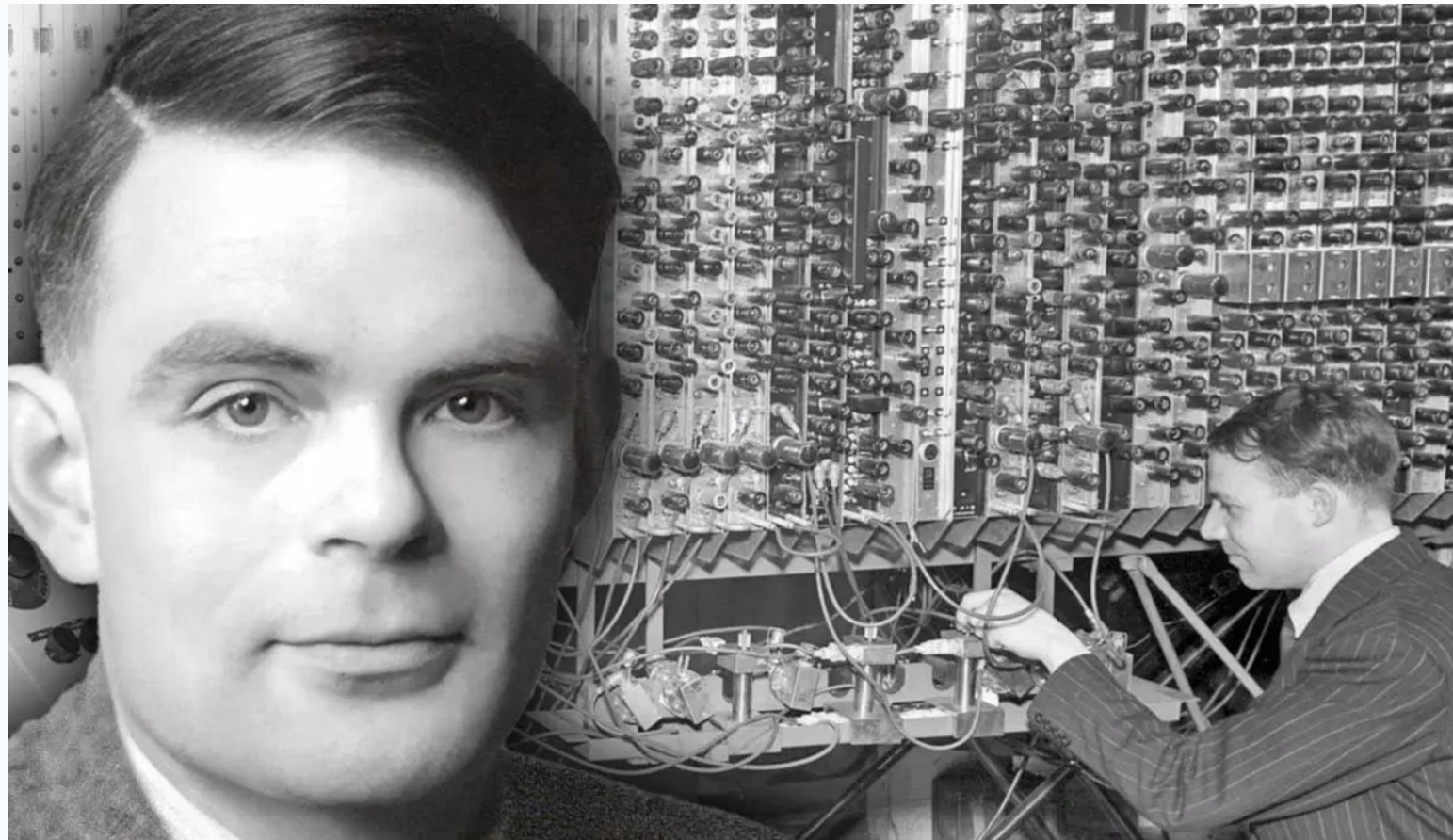
- **1950:** Alan Turing published "Computing Machinery and Intelligence"
- Turing asked the question, "Can machines think?", which later become known as the "Turing Test"



# A Review of AI History



# A Review of AI History



# Birth of AI

- 1955: The term "artificial intelligence" is coined in a proposal submitted by **John McCarthy** (Dartmouth College), **Marvin Minsky** (Harvard University), **Nathaniel Rochester** (IBM), and **Claude Shannon** (Bell Lab)

*"The study is to proceed on the basis of the conjecture that **every aspect of learning** or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it."*

- 1956: The workshop at Dartmouth College was regarded as the official birthplace of AI





Nathaniel Rochester

Marvin Minsky

John McCarthy

Ray Solomonoff

Peter M. Milner

Claude Shannon

A PROPOSAL FOR THE  
DARTMOUTH SUMMER RESEARCH  
ON ARTIFICIAL INTELLIGENCE

J. McCarthy, Dartmouth  
M. L. Minsky, Harvard  
N. Rochester, I. B. M.  
C. E. Shannon, Bell Tel

IN THIS BUILDING DURING THE SUMMER  
JOHN MCCARTHY (DARTMOUTH COLLEGE), MARVIN  
NATHANIEL ROCHESTER (IBM), AND CLAUDE SHANNON  
CONDUCTED

THE DARTMOUTH SUMMER RESEARCH  
ON ARTIFICIAL INTELLIGENCE

FIRST USE OF THE TERM "ARTIFICIAL INTELLIGENCE"  
FOUNDING OF ARTIFICIAL INTELLIGENCE AS A RESEARCH FIELD

"To proceed on the basis of the conjecture  
that every aspect of learning or any other feature  
can in principle be so precisely described that a machine can be made to do it"

IN COMMEMORATION OF THE PROJECT'S 50th ANNIVERSARY  
JULY 13, 2006

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.

The following are some aspects of the artificial intelligence problem:

### **1 Automatic Computers**

If a machine can do a job, then an automatic calculator can be programmed to simulate the machine. The speeds and memory capacities of present computers may be insufficient to simulate many of the higher functions of the human brain, but the major obstacle is not lack of machine capacity, but our inability to write programs taking full advantage of what we have.

### **2. How Can a Computer be Programmed to Use a Language**

It may be speculated that a large part of human thought consists of manipulating words according to rules of reasoning and rules of conjecture. From this point of view, forming a generalization consists of admitting a new word and some rules whereby sentences containing it imply and are implied by others. This idea has never been very precisely formulated nor have examples been worked out.

### **3. Neuron Nets**

How can a set of (hypothetical) neurons be arranged so as to form concepts. Considerable theoretical and experimental work has been done on this problem by Uttley, Rashevsky and his group, Farley and Clark, Pitts and McCulloch, Minsky, Rochester and Holland, and others. Partial results have been obtained but the problem needs more theoretical work.

The Rockefeller Foundation is being asked to provide financial support for the project on the following basis:

- 1. Salaries of \$1200 for each faculty level participant who is not being supported by his own organization. It is expected, for example, that the participants from Bell Laboratories and IBM Corporation will be supported by these organizations while those from Dartmouth and Harvard will require foundation support.
- 2. Salaries of \$700 for up to two graduate students.
- 3. Railway fare for participants coming from a distance.
- 4. Rent for people who are simultaneously renting elsewhere.
- 5. Secretarial expenses of \$650, \$500 for a secretary and \$150 for duplicating expenses.

6. Organization expenses of \$200. (Includes expense of reproducing preliminary work by participants and travel necessary for organization purposes.

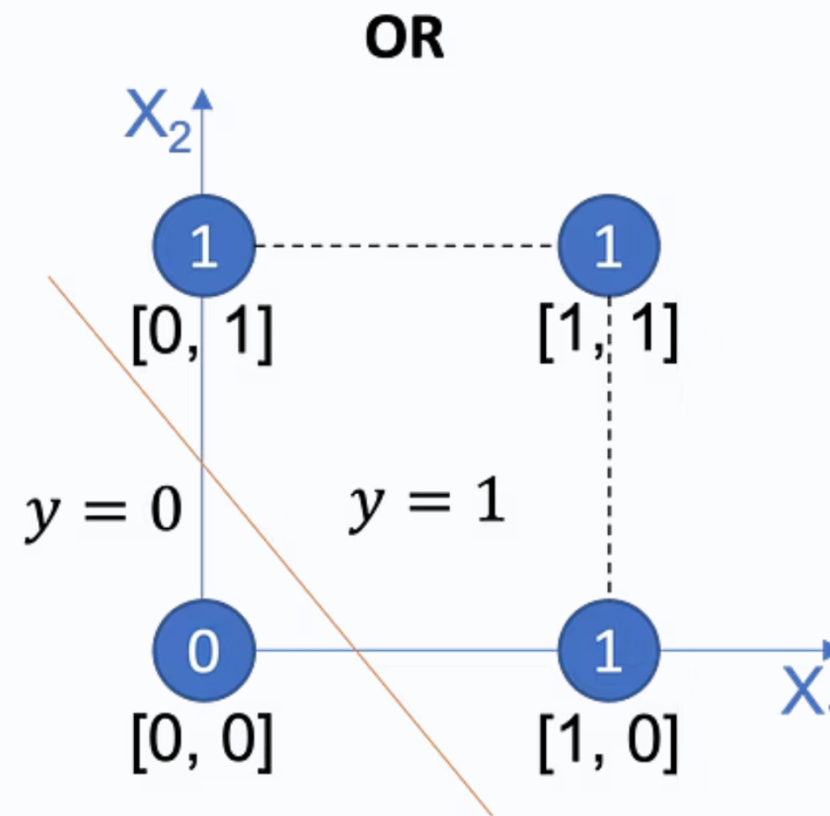
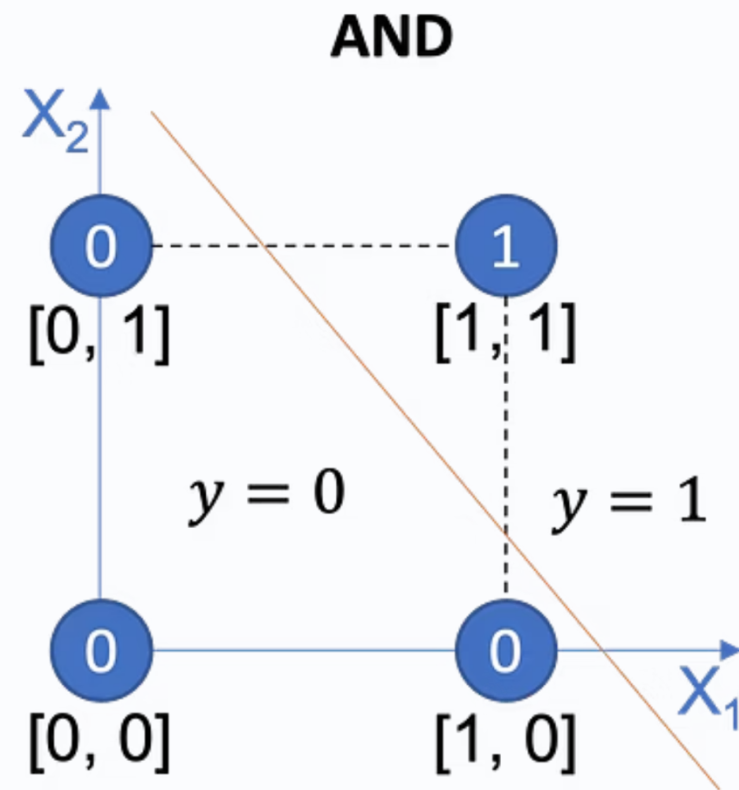
7. Expenses for two or three people visiting for a short time.

Estimated Expenses

|                                             |          |
|---------------------------------------------|----------|
| 6 salaries of 1200                          | \$7200   |
| 2 salaries of 700                           | 1400     |
| 8 traveling and rent expenses averaging 300 | 2400     |
| Secretarial and organizational expense      | 850      |
| Additional traveling expenses               | 600      |
| Contingencies                               | 550      |
|                                             | -----    |
|                                             | \$13,500 |

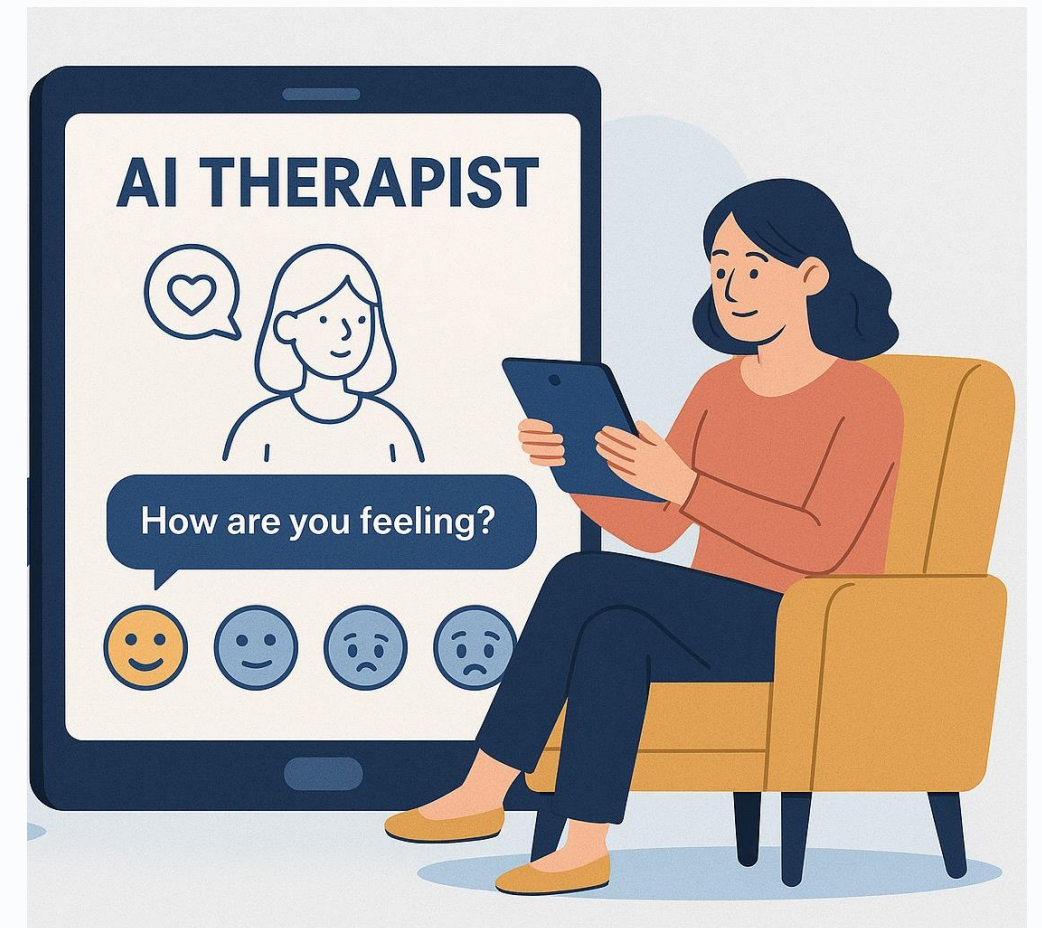
# A Review of AI History

- 1957: Frank Rosenblatt develops the Perceptron, an early artificial neural network enabling pattern recognition based on a two-layer learning network



# A Review of AI History

- 1965: Joseph Weizenbaum develops **ELIZA**, an interactive program that carries on a dialogue in English language on any topic
- Some people attributed human-like feelings to the computer program



Welcome to

|        |        |      |        |         |
|--------|--------|------|--------|---------|
| EEEEEE | LL     | IIII | ZZZZZZ | AAAAA   |
| EE     | LL     | II   | ZZ     | AA AA   |
| EEEEEE | LL     | II   | ZZZ    | AAAAAAA |
| EE     | LL     | II   | ZZ     | AA AA   |
| EEEEEE | LLLLLL | IIII | ZZZZZZ | AA AA   |

Eliza is a mock Rogerian psychotherapist.  
The original program was described by Joseph Weizenbaum in 1966.  
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?  
YOU: Men are all alike.  
ELIZA: What is the connection, do you suppose ?  
YOU: They're always bugging us about something or other.  
ELIZA: Can you think of a specific example ?  
YOU: Well, my boyfriend made me come here.  
ELIZA: Is it important to you that your boyfriend made you come here ?  
YOU: He says I'm depressed much of the time.  
ELIZA: I am sorry to hear that you are depressed.  
YOU: It's true. I am unhappy.  
ELIZA: Can you explain what made you unhappy ?  
YOU:

# Overwhelming Optimism

- The New York Times reported the **Perceptron** to be the embryo of an electronic computer that [the Navy] expects will be able to walk, talk, see...
- The New Yorker called it a "remarkable machine... capable of what amounts to thought"

## NEW NAVY DEVICE LEARNS BY DOING

Psychologist Shows Embryo  
of Computer Designed to  
Read and Grow Wiser

WASHINGTON, July 7 (UPI)

—The Navy revealed the embryo of an electronic computer today that it expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence.

The embryo—the Weather Bureau's \$2,000,000 "704" computer—learned to differentiate between right and left after fifty attempts in the Navy's demonstration for newsmen.

The service said it would use this principle to build the first of its Perceptron thinking machines that will be able to read and write. It is expected to be finished in about a year at a cost of \$100,000.

Dr. Frank Rosenblatt, designer of the Perceptron, conducted the demonstration. He said the machine would be the first device to think as the human brain. As do human be-

ings, Perceptron will make mistakes at first, but will grow wiser as it gains experience, he said.

Dr. Rosenblatt, a research psychologist at the Cornell Aeronautical Laboratory, Buffalo, said Perceptrons might be fired to the planets as mechanical space explorers.

### Without Human Controls

The Navy said the perceptron would be the first non-living mechanism "capable of receiving, recognizing and identifying its surroundings without any human training or control."

The "brain" is designed to remember images and information it has perceived itself. Ordinary computers remember only what is fed into them on punch cards or magnetic tape.

Later Perceptrons will be able to recognize people and call out their names and instantly translate speech in one language to speech or writing in another language, it was predicted.

Mr. Rosenblatt said in principle it would be possible to build brains that could reproduce themselves on an assembly line and which would be conscious of their existence.

1958 New York  
Times...

In today's demonstration, the "704" was fed two cards, one with squares marked on the left side and the other with squares on the right side.

### Learns by Doing

In the first fifty trials, the machine made no distinction between them. It then started registering a "Q" for the left squares and "O" for the right squares.

Dr. Rosenblatt said he could explain why the machine learned only in highly technical terms. But he said the computer had undergone a "self-induced change in the wiring diagram."

The first Perceptron will have about 1,000 electronic "association cells" receiving electrical impulses from an eye-like scanning device with 400 photo-cells. The human brain has 10,000,000,000 responsive cells, including 100,000,000 connections with the eyes.

# Overwhelming Optimism

- With the initial success, it was a time of high optimism, with all leaders of the field predicting that AI would be "solved" in a matter of years...
  - **Herbert Simon**: Machines will be capable, within **twenty years**, of doing any work a man can do
  - **Marvin Minsky**: Within **10 years**, the problems of artificial intelligence will be substantially solved
  - **Claude Shannon**: I visualize a time when we will be to robots what dogs are to humans, and I'm rooting for the machines

# Underwhelming Results

- 1966: The Automatic Language Processing Advisory Committee (ALPAC) report gained notoriety for being **very skeptical** of **research done in machine translation**
- The U.S. government to reduce funding dramatically

*The spirit is willing but the flesh is weak.*



(Russian)



*The vodka is good but the meat is rotten.*

# The First AI Winter

- ① The collapse of generalized AI triggered the first brutal AI winter.

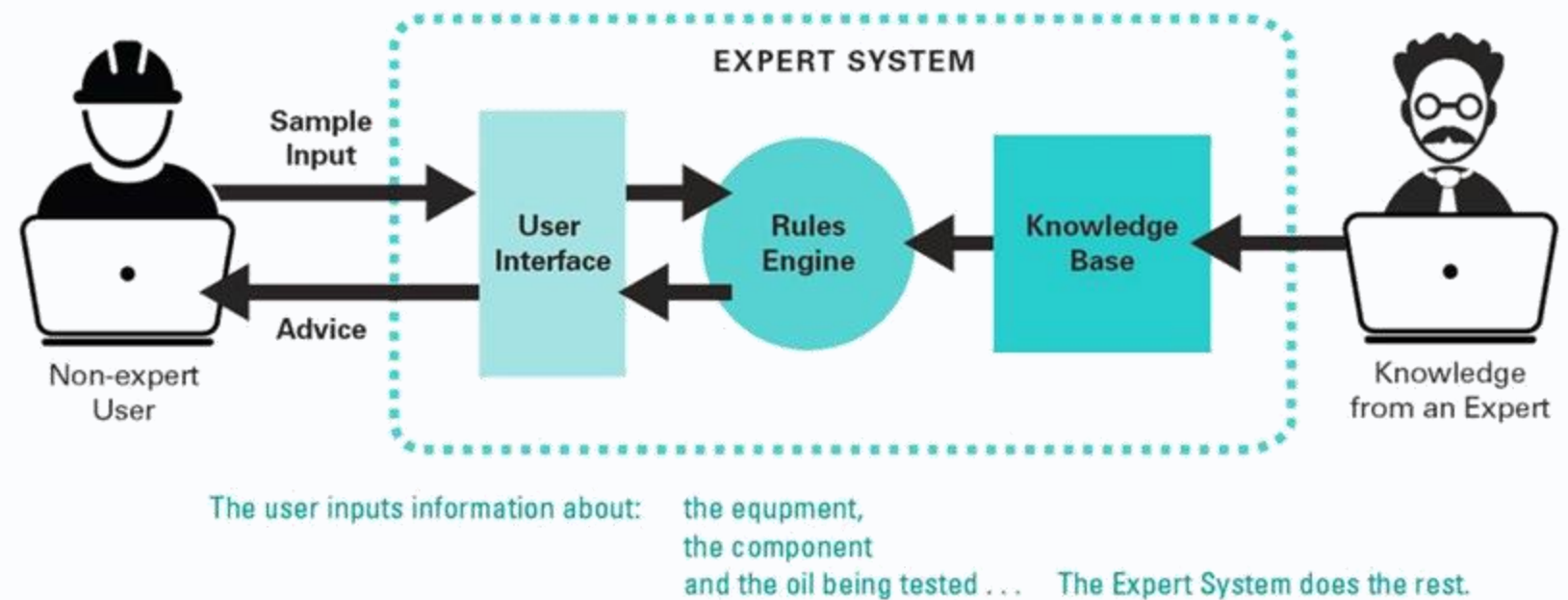
# Boom 2 (Late 1970s– Late 80s): Can Machines Become Experts in Narrow Domains?

- Main Goal
  - Use computers to mimic human experts (doctors, engineers).
- What AI Could Do
  - Rule-based diagnosis (e.g., MYCIN for infections)
  - Configure systems (e.g., XCON for computer assembly)
- Key Technologies
  - Expert Systems (if-then rules), Symbolic AI
- Major Issue
  - Rule explosion. Real-life is messy, rules conflicted, and systems were fragile.
- What Failed
  - Couldn't adapt to new situations or uncertainty
  - Cost of updating knowledge was too high → Second AI Winter

# Expert Systems

**Expert Systems (1970-80s):** instead of the solve-it-all optimism from the 1950-60s, researchers focused on building narrow practical systems in targeted domains

- **MYCIN:** diagnosed bacterial infections and recommended antibiotics
- **DENDRAL:** assisted chemists in identifying unknown organic molecules



# Expert Systems

- **Limitation**
  - They can't adapt: fixed rules can't handle the uncertainty of the real world
  - Updating rules required ongoing input from human experts
  - Rules quickly became too complex to create and maintain
  - Difficult to scale

# A Review of AI History

- **1986: D. Rumelhart, G. Hinton, and R. Williams** published "Learning representations by back-propagating errors"
- They precisely explained how to use the **back-propagation** to train multi-layer perceptrons

nature

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[nature](#) > [letters](#) > article

Letter | Published: 09 October 1986

## Learning representations by back-propagating errors

[David E. Rumelhart, Geoffrey E. Hinton & Ronald J. Williams](#)

[Nature](#) **323**, 533–536 (1986) | [Cite this article](#)

217k Accesses | 23k Citations | 762 Altmetric | [Metrics](#)

### Abstract

We describe a new learning procedure, back-propagation, for networks of neurone-like units. The procedure repeatedly adjusts the weights of the connections in the network so as to minimize a measure of the difference between the actual output vector of the net and the desired output vector. As a result of the weight adjustments, internal 'hidden' units which are not part of the input or output come to represent important features of the task domain, and the regularities in the task are captured by the interactions of these units. The ability to create useful new features distinguishes back-propagation from earlier, simpler methods such as the perceptron-convergence procedure<sup>1</sup>.

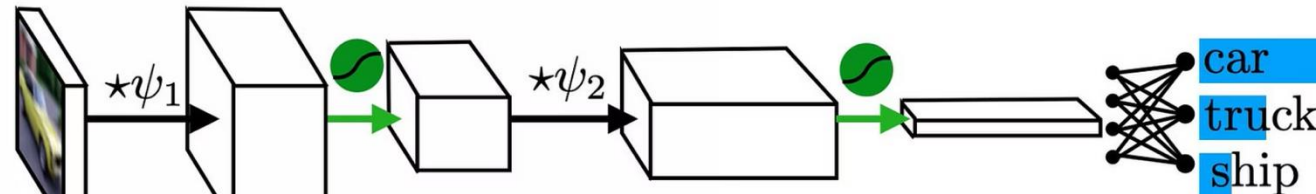
# A Review of AI History

- 1989: **Yann LeCun** at AT&T Bell Labs successfully applied a backpropagation algorithm to a multi-layer neural network, recognizing handwritten ZIP codes

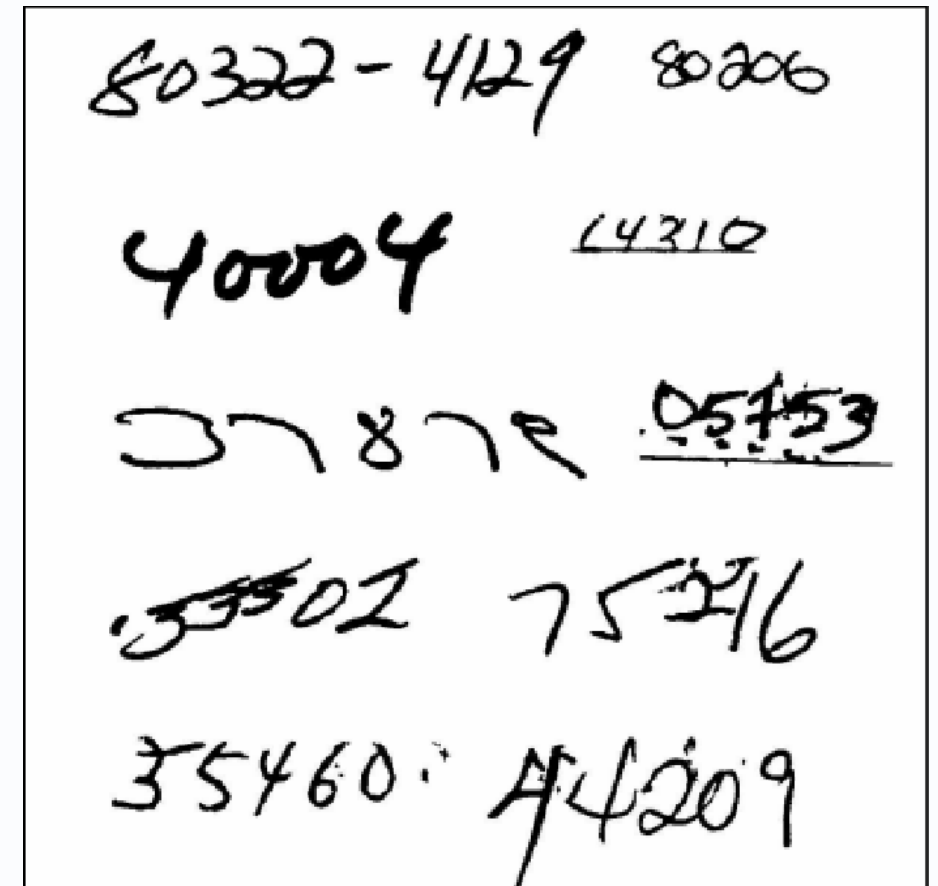
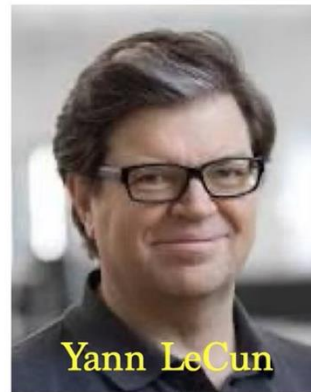
## Backpropagation Applied to Handwritten Zip Code Recognition

Y. LeCun  
B. Boser  
J. S. Denker  
D. Henderson  
R. E. Howard  
W. Hubbard  
L. D. Jackel

AT&T Bell Laboratories Holmdel, NJ 07733 USA



The ability of learning networks to generalize can be greatly enhanced by providing constraints from the task domain. This paper demonstrates how such constraints can be integrated into a backpropagation network through the architecture of the network. This approach has been successfully applied to the recognition of handwritten zip code digits provided by the U.S. Postal Service. A single network learns the entire recognition operation, going from the normalized image of the character to the final classification.

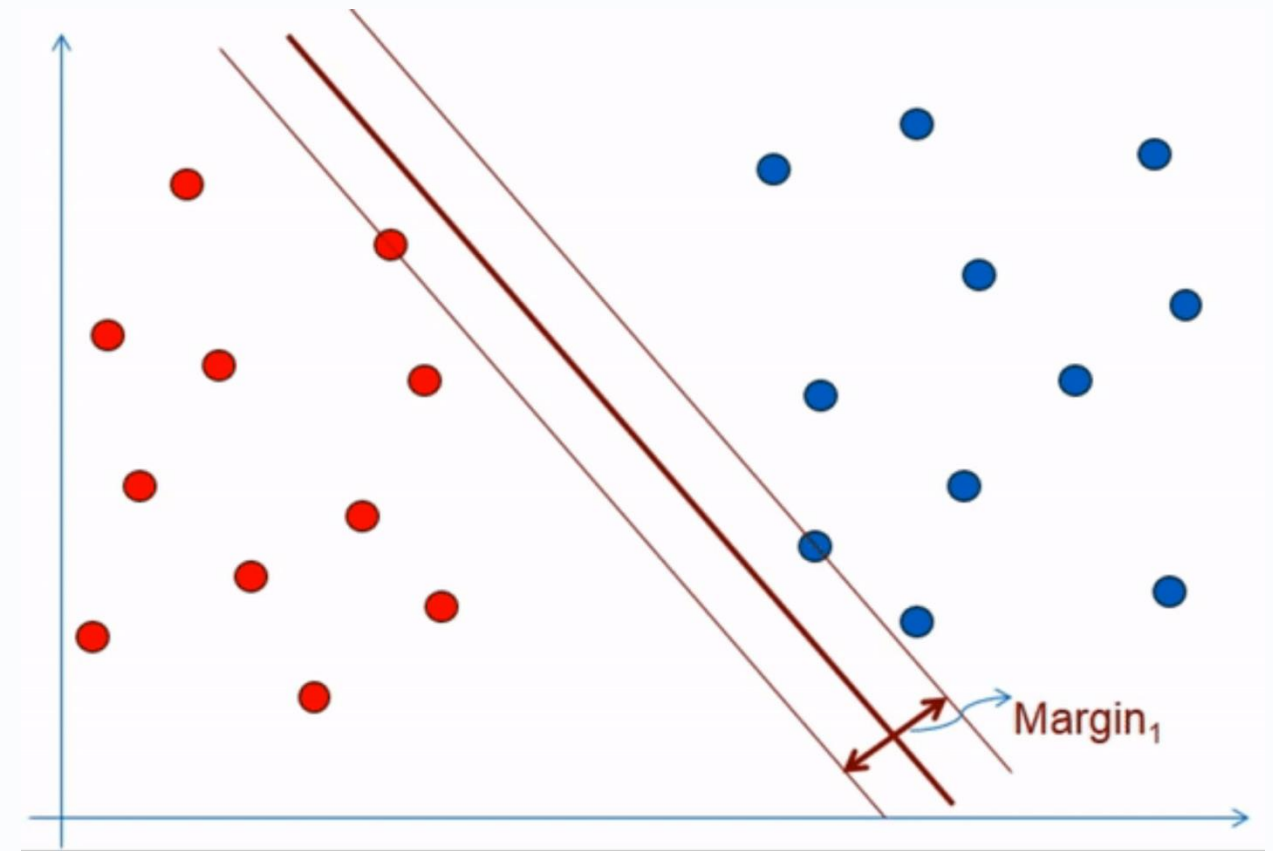


# The Second AI Winter

Even with new advancements, from early 1990s, commercial systems were phased out; enthusiasm in AI faded; funding dried up

# A Review of AI History

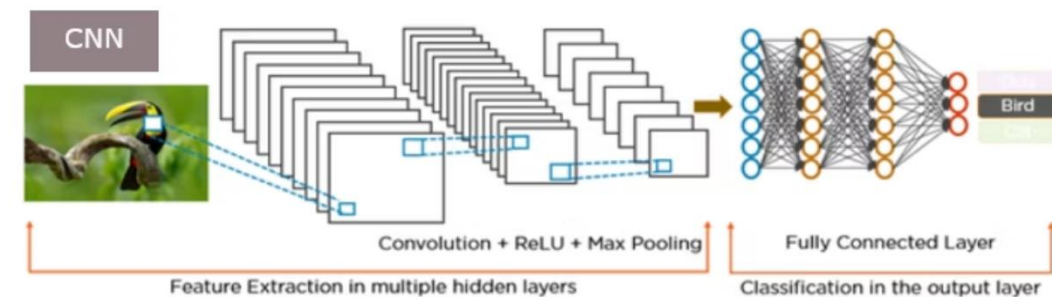
- 1995: C. Cortes and V. Vapnik introduced Support Vector Machines (SVM), which is a powerful method for classification and regression



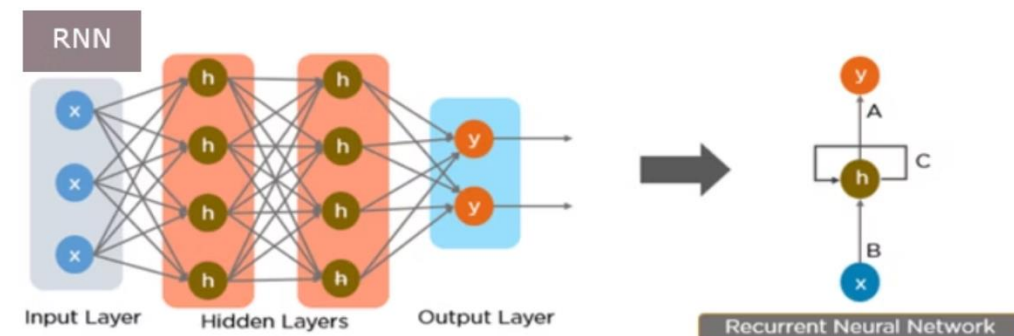
# A Review of AI History

- **1997: S. Hochreiter and J. Schmidhuber** propose Long Short-Term Memory (**LSTM**), a type of a recurrent neural network used in handwriting recognition and speech recognition

Convolutional Neural Network



Recurrent Neural Network



# A Review of AI History

## Deep Blue

The first computer chess-playing program to beat a reigning world chess champion (1997).



# Boom3 after 2000s: Can Machines Learn From Data or Do more?

- Main Goal

Let machines learn patterns from data, not rules from humans.

Machines that can see, hear, and understand.

- What AI Could Do

- Recognize images (e.g., cats, faces)
  - ImageNet/AlexNet revolution
- Translate languages decently (Google Translate)
- Beat humans at Go (AlphaGo)
- Understand spoken language (Siri, Alexa)

- Key Technologies

Deep learning (CNNs, RNNs, reinforcement learning)

...

- Major Issue

Needs huge data and computing power. Still struggles with reasoning and generalization.

- What Changed

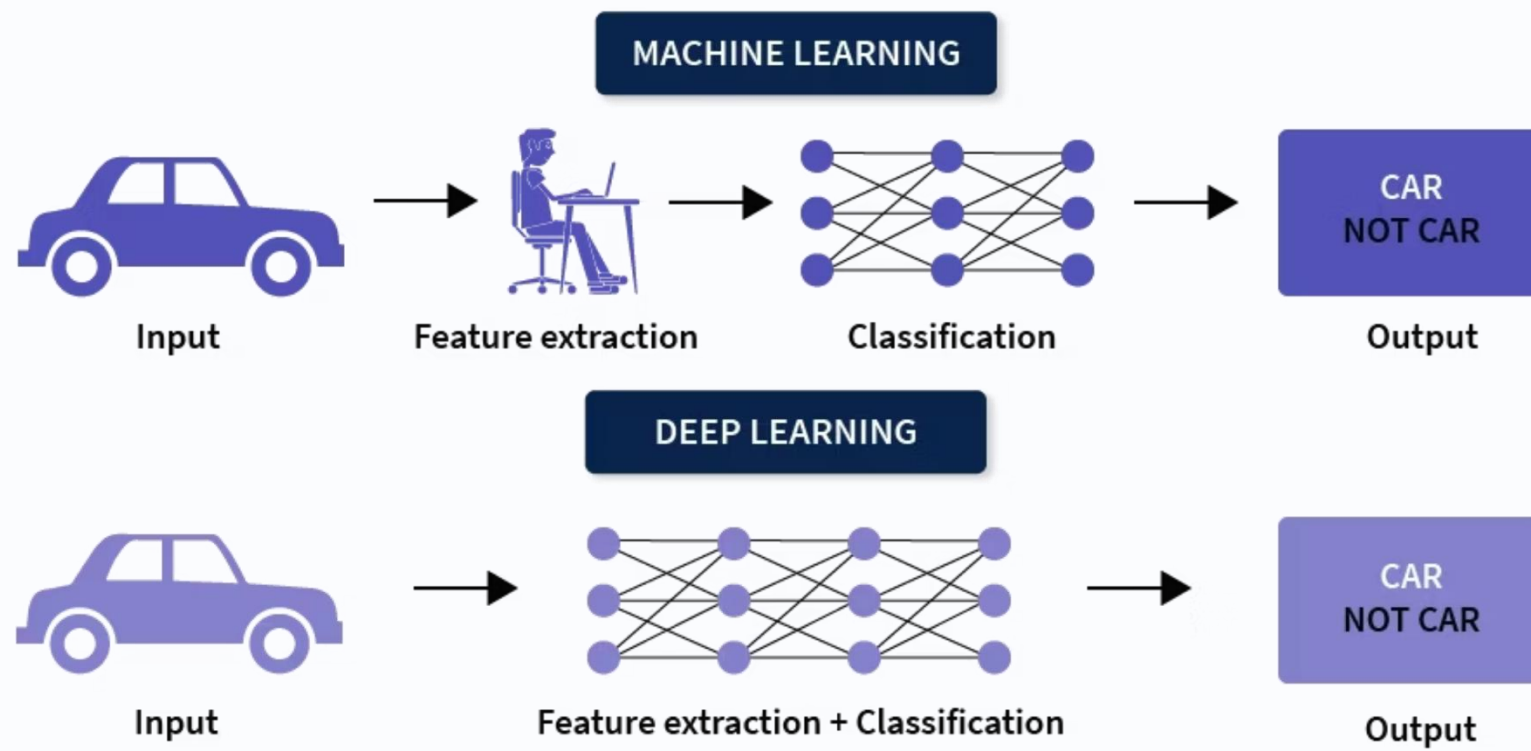
- Learning from massive datasets became possible
- Neural networks finally beat traditional models
- AI felt "real" to the public for the first time

# Why New Boom?

- Availability of powerful computer hardware
- Collection of immense data sets
- Application of solid mathematical methods
- New method

# A Review of AI History

- 2006: **Geoffrey Hinton** publishes "Learning Multiple Layers of Representation," summarizing the ideas that have led to "multilayer neural networks and training them to generate sensory data rather than to classify it", i.e., **deep learning**



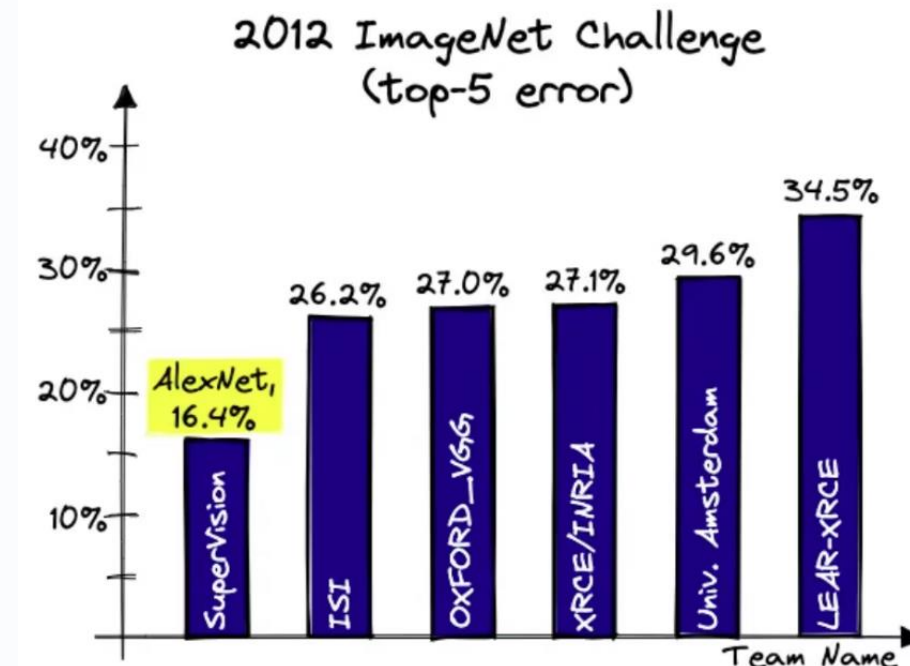
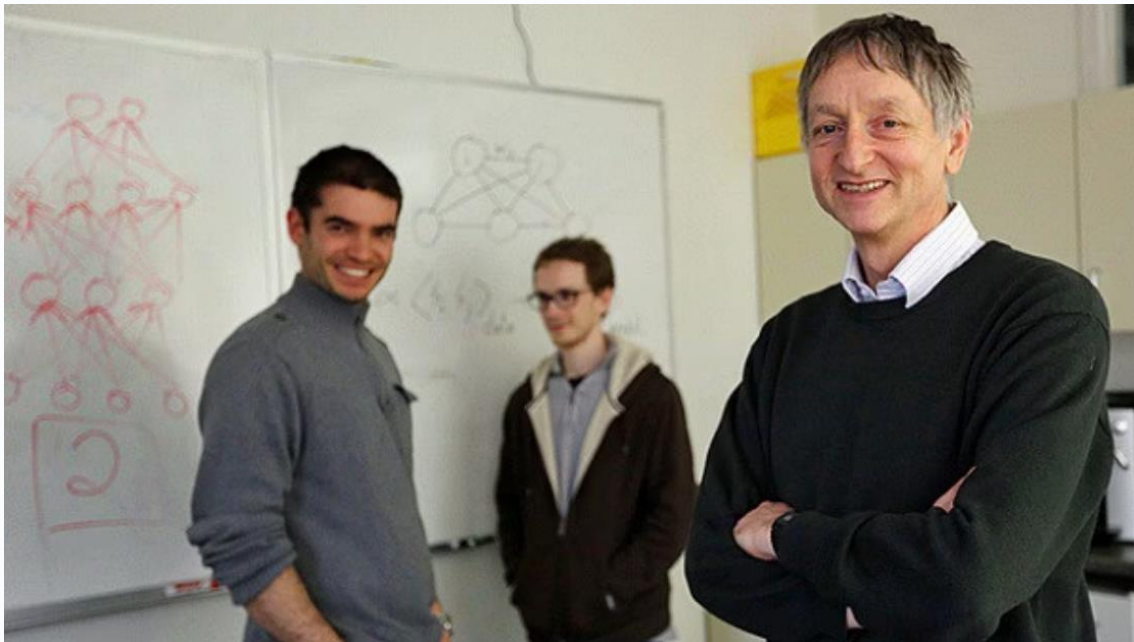
# A Review of AI History

- 2007: **Fei-Fei Li** started to assemble **ImageNet**, which is one of the largest datasets for image recognition, containing millions of labeled images across thousands of categories, providing researchers with a benchmark for training and evaluating machine learning models



# A Review of AI History

- **2012**: Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton designed a convolutional neural network (**AlexNet**) that achieved an error rate of 16% in the ImageNet Large Scale Visual Recognition Challenge, **a significant improvement over the error rate** achieved by the best entry the year before



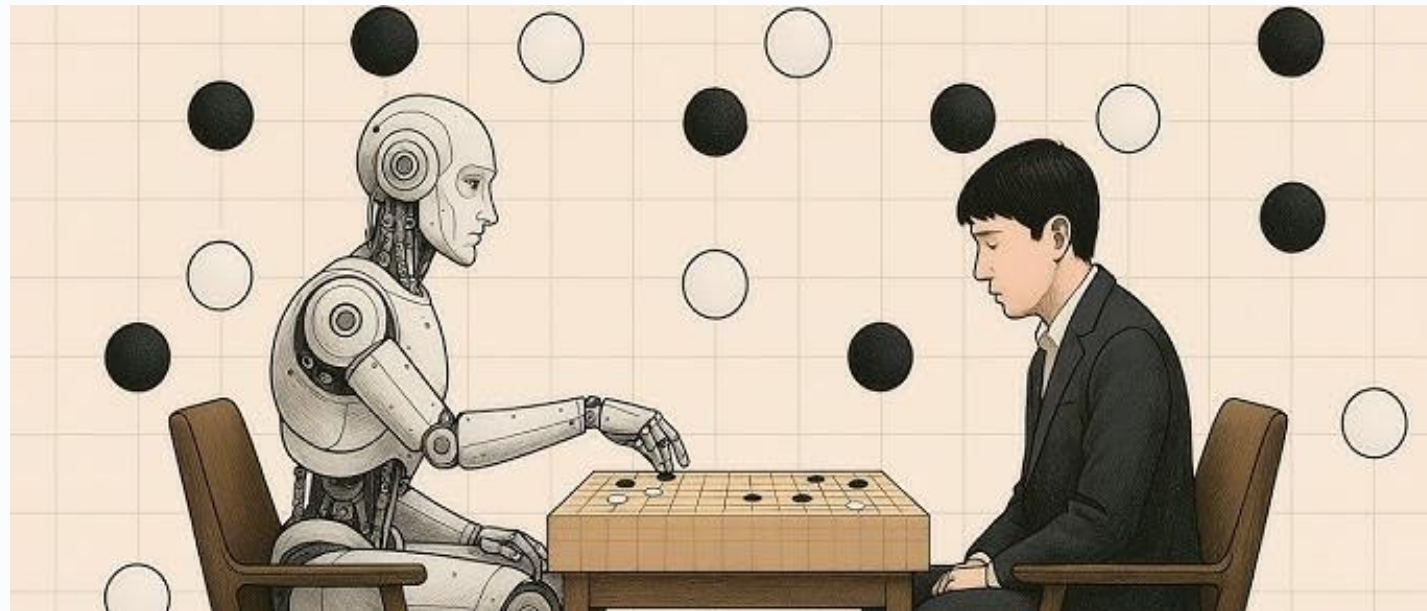
# A Review of AI History

- 2015: Kaiming He's **ResNet** enabled a leap from ~20 layers to 1000+ layers, revolutionizing the scalability of neural networks (K. He is a CUHK alumnus of the IE PhD program in 2011)



# A Review of AI History

- 2016: AlphaGo is based on deep **reinforcement learning** and defeats Lee Sedol
- The ancient game of Go is considered straightforward to learn but **incredibly difficult**—bordering on impossible—for any computer system to play, given the vast number of potential positions



# A Review of AI History



- [https://www.bilibili.com/video/BV13x411x7kY/?spm\\_id\\_from=333.337.search-card.all.click&vd\\_source=b0e15bdbda6a17abcc8c653915e36a8f](https://www.bilibili.com/video/BV13x411x7kY/?spm_id_from=333.337.search-card.all.click&vd_source=b0e15bdbda6a17abcc8c653915e36a8f)

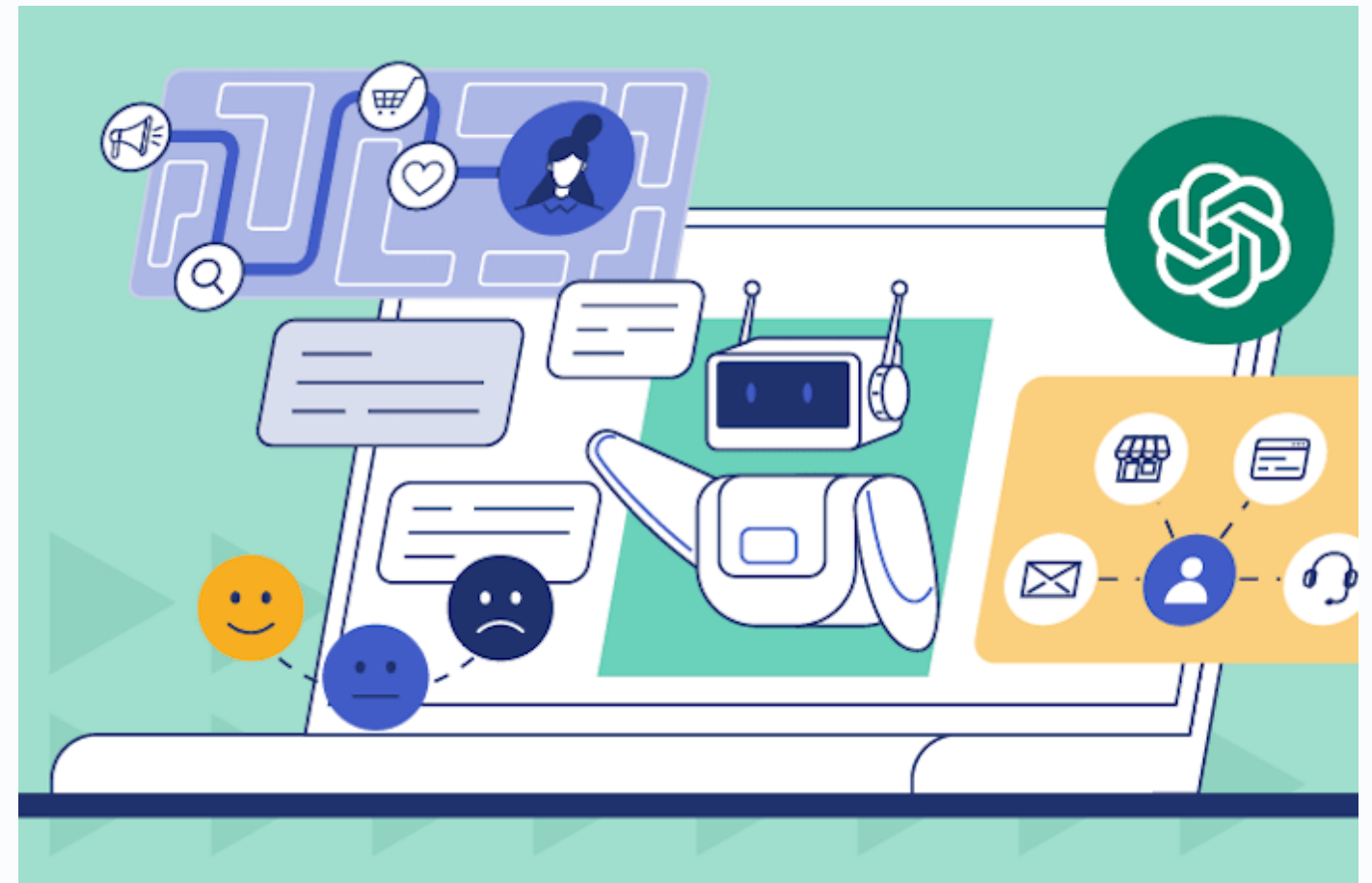
# A Review of AI History

- 2017: Researchers at Google published the paper **Attention Is All You Need**
- Focuses on patterns of attention, essentially teaching AI to decide **which words or pieces of information matter most in context**. This approach
- **Enabled the efficient handling and understanding of immense amounts of text**, transforming language-based applications and laying the foundation for large language models.
- Example: “On Friday, the judge issued a sentence”

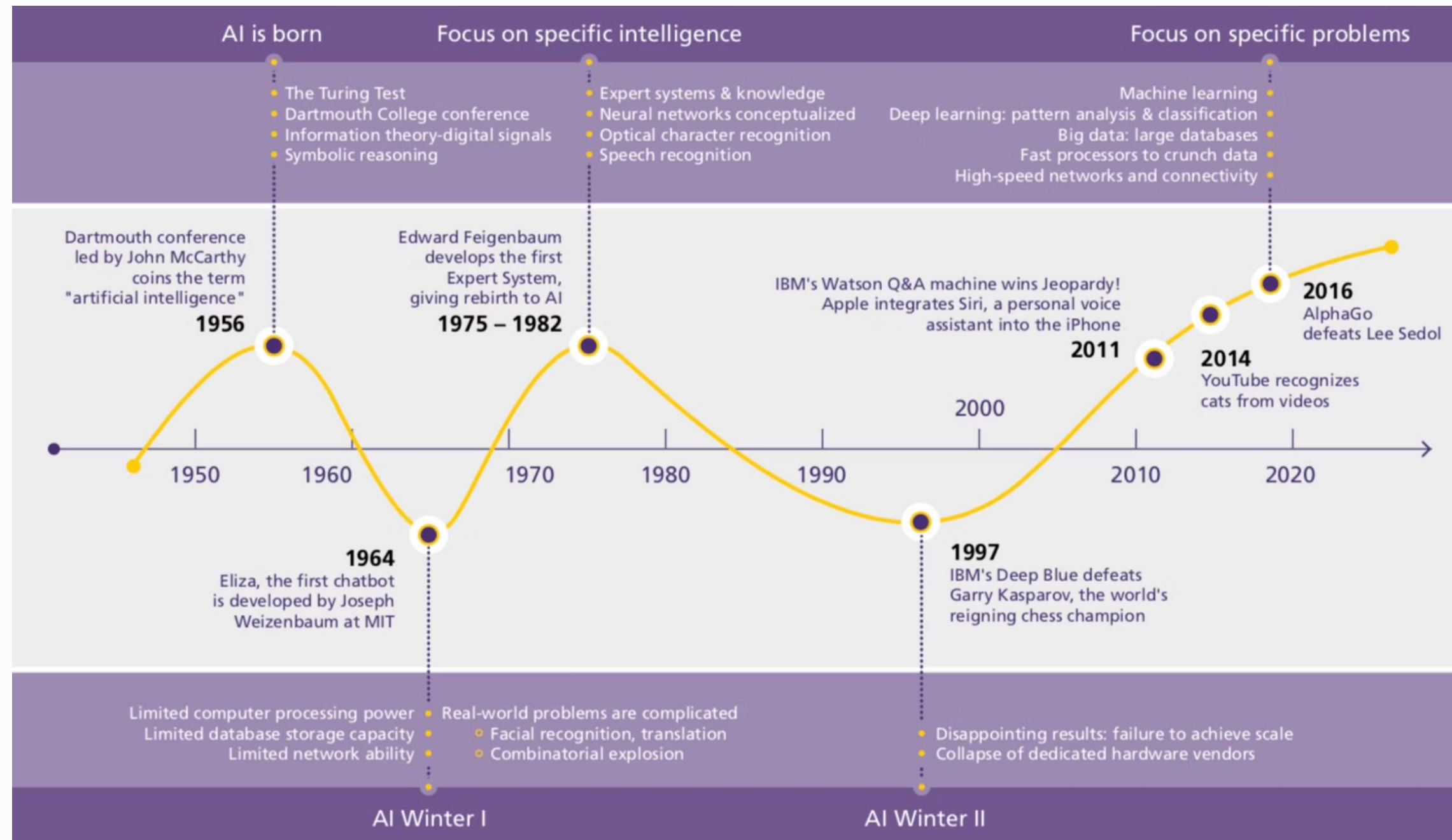


# A Review of AI History

- **2022:** ChatGPT (by OpenAI)
- Based on the GPT (Generative Pre-trained Transformer)
- Represent a major leap in AI-powered conversational agents



# Summary



Any limitation in current AI?

Are there new directions?

# Sources

- 114 Milestones In The History Of Artificial Intelligence: <https://www.forbes.com/sites/gilpress/2021/05/19/114-milestones-in-the-history-of-artificial-intelligence-ai/>
- General: AI history: <https://stanford-cs221.github.io/autumn2024-extra/modules/general/history.pdf>
- Proclaimed Dead Twice: The AI Winters: <https://medium.com/@bergholz/proclaimed-dead-twice-the-ai-winters-5848c7f8a29f>
- Presentations, Tutorials, and Workshops by Professor Yi Ma: <https://people.eecs.berkeley.edu/~yima/Talks.html>
- Presentations, Tutorials, and Workshops by Professor Yi Ma: <https://people.eecs.berkeley.edu/~yima/Talks.html>

Questions?

# Course Feedback

